Gate B v2 and metrics: change log

What changed, why, what the re-run showed, and what the dashboard does and does not show yet.

> **Where this stands.** Phases 0 to 4 are done and committed on the branch gate-b-v2-and-metrics-consolidation. It is not pushed and not merged. **Metrics v2 is live on that branch. Gate v2 is not:** the Gate B tab still runs v1 until you approve the thresholds. Phases 5 and 6 are waiting on that.
>
> **The re-run:** 20261105-132328-ab35, same approved card, on the new code. Daily returns, trades and holdings are byte-identical to the old run, so the backtest did not change. Only how results are measured and shown changed.

Commits

Commit	What	Why
925289c	Freeze Gate B v1: version constants, manifest and workbook fields, baseline for every past run	So every later result can be compared with what v1 really said. All 12 past runs that reached a result are recorded, and all 12 failed v1.
5139c54	One metrics module: every dashboard, workbook, report and gate reads ros/metrics/core.py	The same metric name gave different numbers depending on where you looked.
fab1b88	Gate v2: hard blocks, evidence scorecard, PASS/FAIL/INCOMPLETE, mandate file, card fields	Built so it could be simulated. Not wired into any run, and its thresholds are marked as candidates.
c471338	Calibrate Gate B by simulation: 8,000 strategies, real-data references, power curves and a proposal	So thresholds come from simulation and not from any one strategy. Clenow was not used.
b26130c	Fixes from the dashboard check: first-day trades count in turnover, older runs label their old Sharpe, runs stop copying core metrics	The dashboard check found these on the re-run.

50 files changed, 3827 insertions(+), 365 deletions(-), compared with the tag gate-v1. Tests: 636 passed, 1 skipped on the final commit (570 passed before this work). ruff: 182 errors, down from 185, none new.

Metric definitions: before and after

Metric	Before	Now	Why
Sharpe	Four formulas: (CAGR - 6%) / vol on the tile, (CAGR - compounded cash) / vol in tables, (mean x 252 - 6%) / vol in Excel, and mean excess / stdev in the deflated Sharpe	One: mean(r - c) / stdev(r - c) x sqrt(252)	One strategy showed several Sharpes under one name
Cash rate	6% typed in the tile and workbook, 5% in the Excel tearsheet	Read from the card. Blank with a warning if the card has none	No hard-coded rates
Max drawdown	Positive in some places, negative in others	Daily values, positive	One sign
Drawdown duration	Calendar days	Trading days	As you specified
Tracking error	Sample in one path, population in another	Sample stdev of daily gaps	One definition
Up / down capture	Daily means in one path, monthly in another	**Monthly means**. My choice: your brief named no frequency	One definition
95% CVaR	Rolling 21-day in one path, daily and negative in another	**Daily, positive loss**. My choice of frequency	As you specified the sign
Turnover	Mean x 252 in one path, sum / years in another	Sum / years	One definition
Bootstrap	Geometric Sharpe, slow loop	The one Sharpe, same settings (2,000 draws, block 21, seed 7), plus a paired drawdown gap as evidence only	You kept the method and asked for the one Sharpe

Added

What	Where	Why
Core performance table: the same 16 metrics on every run	Results tab; Core metrics sheet in Excel	Each run's own tables differ. Sortino and Calmar are now on the page
Net of the house cost beside gross	Results tab, Excel. Cost from config/costs.yaml	Your rule: always report gross and net
Capacity: trade size against 20-day traded value	Code done, shows as not computed	Needs your AUM in config/fund.yaml
Post-publication metrics	Results tab, Excel	Separates the paper's own period from what came after
Price-index label and a total-return hook	Results tab, Excel	A price benchmark flatters relative figures
Diagnostics section	Results tab	Rows tied to one strategy's rules are not core performance
gate_version and metrics_version	manifest.json, workbook header	Every result says what produced it
Card fields: n_configs_selected_from, n_sensitivities, rules_fixed_in_print, fits_parameters	ros/cards/schema.py	Versions chosen between must not be confused with robustness checks
Config files: costs, fund, mandate, gate thresholds	config/	Settings are read, not typed. Blank fund values stay blank

Cut, kept and left alone

Item	Status
Mean drawdown, vol consistency, hit rate, monthly CVaR, the geometric and duplicate Sharpes	**Cut** from the pipeline report and metrics modules
Sterling, Treynor, Omega, Gain to Pain, tail ratio, adjusted beta, Jensen's alpha, extreme capture, Ret +Ve/-Ve, positive/negative volatility, average annual drawdown, 99% VaR/CVaR	**Still in the code, unused by Lightyear.** They live only in tearsheet.py and excel_tearsheet.py, which you said not to delete until you confirm
Who uses the tearsheet code	scripts/universal_backtester_momentum_rotation.py and three test files. CLAUDE.md mentions it. An old Lee-Swaminathan run log printed one, with no script left in the repo
Geometric Sharpe	**Kept in one place on purpose** (ros/metrics/legacy_v1.py): a paper's printed Sharpe has to be compared on its own definition. Also used to simulate v1
validate/independent.py	**Kept separate on purpose:** it is the clean-room cross-check, and its value is that it imports nothing from ros
universal_backtester/validation.py	**Not consolidated.** Its own bootstrap and deflated Sharpe functions remain, but no Lightyear path or gate uses them

Gate v2 and the calibration

Gate v2 has hard blocks, a scorecard and a status of PASS, FAIL or INCOMPLETE. In 8,000 simulated strategies, the proposed thresholds gave at most 2.2% false passes and passed 91.8% of strategies with a true excess Sharpe of 1.0. 60% of pure-factor strategies get through the hard blocks alone. Full results: gate-b-v2-calibration.pdf. **The proposal was committed in c471338 before this re-run, so Clenow played no part in it.**

The re-run against the last run

Check	Result
Daily returns, trades, holdings, comparators	Byte-identical to the old run
Card	Same sha256 (4b0d6260e6ac). All 28 items agree, and all 18 rule checks pass
Evidence supports (written by Claude)	OBSERVE before, OBSERVE now
Workbook	Recalculated: 71,148 formulas, 0 errors

Gate B criterion (v1)	Old run	Re-run	Result
deflated Sharpe clears selection bias	0.00	0.00	fail
positive Sharpe in every out-of-sample window	-0.29	-0.24	fail
advantage over benchmark is significant	0.058	0.054	fail
differentiated from existing book	no comparison	no comparison	not computed
improves the book's information ratio	no comparison	no comparison	not computed
alpha survives the factor fingerprint	1.52	1.52	fail
implementable within mandate	n/a	n/a	pass
turnover within operational tolerance	1033%	1033%	fail, advisory

> **The v1 gate label needs care here.** The re-run says gate_version v1 because the rules and pass marks are v1's. But criteria 2 and 3 now read the new Sharpe, which is why -0.29 became -0.24 and 0.058 became 0.054. No result changed. The true v1 record is docs/gate_b/gate_v1_baseline.csv.

On the page now	As run (zero cost)	Net at 30bp	NIFTY 500
CAGR	19.80%	17.95%	12.83%
Sharpe	0.83	0.74	0.49
Sortino	1.10	0.97	0.65
Calmar	0.88	0.75	0.34
Maximum drawdown	22.47%	23.99%	38.30%
Drawdown duration (trading days)	547	699	490
Information ratio	0.49	0.36	–

Turnover 10.33 times capital a year. After publication (from 2015-01-01): CAGR 16.42% against 10.92% for the benchmark, Sharpe 0.66 against 0.37.

Dashboard check

I opened the re-run in a real browser and tested each change against the page, manifest and workbook: **39 of 39 checks pass**, including Excel, after recalculation, matching the page on CAGR, volatility, Sharpe, drawdown, Sortino and Calmar. There were no script errors and all charts drew.

Found and fixed by the check

Problem	Cause	Fix
Turnover read 10.33 in one table and 10.26 in another. Net figures slightly too high	**My bug.** The first day's purchases fall the day before the first return, and my code dropped them	Counted on the first day. Now 10.33 in both, and net CAGR 17.95%, matching the full re-simulation
On older runs, the tile said "Sharpe" but showed the old 6% number	My relabel, with their stored numbers unchanged	It now reads "Sharpe (old 6% hurdle)" and says to press Rebuild
"Rolling one-year behind" shows 34.59% in one table and 34.78% in another	A row Claude added in the run uses a 250-day window. The core metric uses 252	Instruction added so future runs do not add their own copy. **This run still shows both**

Not on the dashboard yet

Gap	Why	When
Gate B tab is still v1: eight criteria, no hard blocks, scorecard or INCOMPLETE	Gate v2 is not wired in until you approve thresholds	Phase 5
gate_version is not shown on the page	It is only in the manifest and workbook	Phase 5
Capacity shows not computed. Mandate limits are blank	No AUM or limits supplied	When you supply them
Book tests still show not computed	No book exists: no decision has ever been recorded	When you supply a book
Diagnostics is partial. 4 rows moved, but 7 rule-specific rows Claude added stay in Costs & execution, such as sells on leaving the index and 15%-gap exits	Those rows are written during the run, not from the script's fixed list	Needs a script change: tell me if you want it now
Excel's Validation sheet has no separate Diagnostics section	Only the page was changed	Small fix: Phase 5
Older runs show the new metrics only after Rebuild	Their stored numbers predate v2	Press Rebuild on each
Your usual Lightyear on port 8100 has none of this	It runs the old branch, and it was not running when I last checked	Use port 8101 for now

Decisions needed from you

#	Decision	My recommendation
1	Approve the thresholds in gate-b-v2-calibration.pdf	Approve as proposed
2	Where the deflated Sharpe bar starts	At the benchmark's Sharpe, not zero. The NIFTY 500 itself scores 0.95 otherwise
3	Should anything on the scorecard block?	Yes: deflated Sharpe and factor alpha
4	What counts as the book	The fund's strategies only, not the benchmark
5	AUM, mandate limits, capacity and coverage limits, factor product costs, a book file	Needed from you
6	Delete the tearsheet code?	Archive it, and move or retire the momentum rotation script first
7	Push: the brief says push, but my notes say this repo is public with no pushes	Confirm before Phase 6